Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/13/2005
PI Name:	Forgacs, Gabor Ph.D.		
Project Title:	Comparative Study Of Morphogenesis In 1g And Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	forgacsg@missouri.edu	**Fax:**	FY 573-882-4195
PI Organization Type:	UNIVERSITY	**Phone:**	573-882-3036
Organization Name:	University of Missouri-Columbia		
PI Address 1:	420 Physics Bldg		
PI Address 2:	Department of Physics		
PI Web Page:			
City:	Columbia	**State:**	MO
Zip Code:	65211	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	01/01/2003	**End Date:**	12/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2005 per PI info in 12/2005 (previous end date was 12/31/2006)--1/2009		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1611		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This interdisciplinary study is aimed at the better understanding of early developmental processes both in 1g and reduced gravity. Specifically, we investigate the morphogenetic phenomenon of cell sorting and follow and direct the self-assembly of organ modules of prescribed geometry. Sorting plays crucial role in the final physiological arrangement of tissues and organs. Cell sorting is one of the many self-assembly processes that take place in biological organisms. We are presently developing the technology to control and direct certain self-assembly processes. In particular, we are building tubular organ modules that eventually could be used as vascular or intestinal grafts. For this we use the evolving technology of organ printing, the computer controlled, three dimensional, layer-by-layer deposition of "bioink" particles, uniform-sized hetero- or homo-typical spherical cellular aggregates. The understanding and ability to reproducibly fabricate organ modules will be critical for long space flights and exporting life-form on Earth to other		

	planets. Understanding the effect of reduced gravity on bilogical self-assembly in general and on the early developmental process of cell sorting in particular. Understanding the effect of reduced gravity on early developmental processes will be critical for long space flights and exporting life-form on Earth to other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to engineer organ modules of specifc geometry will have significance for regenerative medicinal purposes, such as organ replacement.
Task Progress:	This project "Comparative studies in 1g and microgravity conditions" has successfully been completed and a manuscript reporting on the results of the research has been submitted to Biophysical Journal. Our project to build organ modules of prescribed geometry using the evolving technology of organ printing has received a big boost from the NSF, which awarded our team a $5 million 5-year grant. The preliminary results that played a crucial role in the successful grant application had been partially generated during the NASA project, as part of studying early developmental phenomena. In particular, in the last NASA grant period we managed to print functional tissue constructs, sheets of synchroneously beating cardiomyocytes. We successfully showed that tissue liquidity, a notion at the basis of the original NASA project, is manifest even at the sub-tissue (i.e. cellular) level.
Bibliography Type:	Description: (Last Updated: 01/23/2009)
Articles in Peer-reviewed Journals	Neagu A, Jakab K, Jamison R, Forgacs G. "The role of physical mechanisms in biological self-organization." Phys Rev Lett. 2005 Oct 21;95(17):178104. Epub 2005 Oct 21. PMID: 16383876 , Oct-2005
Articles in Peer-reviewed Journals	Sun M, Graham JS, Hegedus B, Marga F, Zhang Y, Forgacs G, Grandbois M. "Multiple membrane tethers probed by atomic force microscopy." Biophys J. 2005 Dec;89(6):4320-9. Epub 2005 Sep 23. PMID: 16183875 , Dec-2005
Articles in Peer-reviewed Journals	Hosu BG, Sun M, Marga F, Grandbois M, Forgacs G. "Eukaryotic membrane tethers revisited using magnetic tweezers." Phys Biol. 2007 Jun;4(2):67-78. PMID: 17664652 , http://dx.doi.org/10.1088/1478-3975/4/2/001 , Jun-2007
Articles in Peer-reviewed Journals	Hosu BG, Mullen SF, Critser JK, Forgacs G. "Reversible disassembly of the actin cytoskeleton improves the survival rate and developmental competence of cryopreserved mouse oocytes." PLoS ONE. 2008 Jul 30;3(7):e2787. PMID: 18665248 , Jul-2008
Articles in Peer-reviewed Journals	Hegedüs B, Marga F, Jakab K, Sharpe-Timms KL, Forgacs G. "The interplay of cell-cell and cell-matrix interactions in the invasive properties of brain tumors." Biophys J. 2006 Oct 1;91(7):2708-16. PMID: 16829558 , Oct-2006
Books/Book Chapters	Forgacs G, Newman S. "Biological Physics of the Developing Embryo." Cambridge, UK ; New York : Cambridge University Press, 2005., Nov-2005
Books/Book Chapters	Newman S, Forgacs G. "Complexity and self-organization in development and evolution." in " Complexity in Chemistry, Biology, and Ecology." Ed. Danail Bonchev, Dennis H. Rouvray. New York, NY : Springer, c2005., Dec-2005